



PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
BUMBULI DISTRICT COUNCIL
FORM THREE MIDTERM EXAMINATION



TIME: 3:00HRS

PHYSICS

MARCH 2026

INSTRUCTIONS:

GEPAM science hub

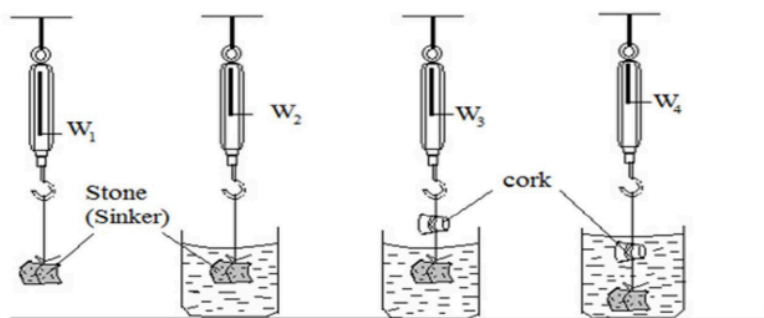
WhatsApp 0748978398

1. This paper contains 10 questions.
2. Answer all questions in section A and B and C.
3. Cell phones are not allowed in the examination room.
4. Non programmable calculator and mathematical table are allowed in the examination room.
5. Where necessary the following constants may be used:
 - Acceleration due to gravity $g = 10 \text{ m/s}^2$
 - Speed of light in air =
 - Speed of sound in air =
 - Pie (π) = 3.14

SECTION A (15 Marks)

Answer all questions in this section

1. i. Which of these resources of energy is non-renewable?
A. Wave energy B. Bio fuels C. Radiant energy D. Fossil fuel E. Geothermal energy
- ii. The acceleration of a moving object may be found from
A. The area under its velocity-time graph. B. The slope of the velocity-time graph.
C. The area under its distance-time graph. D. The slope of the distance-time graph.
E. The slope of the peak of its distance-time graph.
- iii. In a process of charging by induction in static electricity;-
A. A conductor is rubbed with an insulator
B. A charge is produced by friction
C. Negative and positive charges are separated
D. A positive charge induces a positive charge
E. Electrons are sprayed into an object
- iv. The diagram below shows series of experiments performed by a student to determine the relative density of a cork that floats in water. Four reading of the spring balance were obtained.



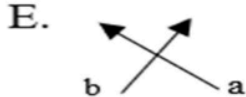
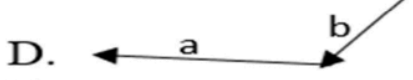
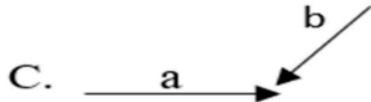
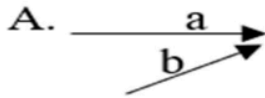
The relative density of the cork is given by:-

A. $\frac{W_1}{W_1 - W_2}$ B. $\frac{W_1 - W_2}{W_3 - W_2}$ C. $\frac{W_3 - W_2}{W_3 - W_4}$ D. $\frac{W_3 - W_1}{W_3 - W_4}$ E. $\frac{W_3}{W_3 - W_4}$

v. In a loading a lorry a man lifts boxes each of weight 100N through a height of 1.5m, if he lifts 4boxes per minute, the average power the man is working is:-

- A. 100 B. 10 C. 600 D. 37.5 E. 2250

vi. Which of the following represent vector $a + b$;



vii. The iris of the eye is similar to the _____ of the lens of camera

- A. Lens
B. Shutters
C. Film
D. Diaphragm
E. View finder

vii. Which of the following apparatus is used for measuring the volume of irregular solid?

- A. Pipette
B. Beaker
C. Measuring cylinder
D. Metre rule
E. Burette

viii. A device that opens or closes a circuit in responses to change in temperature is

- A. Thermometer B. Thermostat C. Switch D. Key E. Thermopile

ix. Which of the phenomenon is taking place when kerosene rises up a wick?

- A. surface tension B. diffusion C. capillarity D. osmosis E. diffusion force

x. The human eyes possess the power of accommodation. This is the power of the eye to

- A. Distinguish between light of different colonies
B. Focus objects at different distances
C. See a series of pictures as a moving picture
D. Decide which of two objects is closer
E. After the diameter of the pupil as the intercity of illumination charges

2. Match the item in the List A with the correct response in the List B, by writing the letter of the most correct response beside the item number in your answer sheet.

LIST A	LIST B
(i) Occur when magnetic material magnetized in an external magnetic field but lose their magnetism when the field is removed.	A. Permanent magnets
(ii) It made from alloy of aluminum, nickel and cobalt that magnetized in external magnetic field and retain or remain with some of the magnetism even after the external field is withdrawn or removed.	B. Magnetization
(iii) Are temporary magnets comes from electric current.	C. Magnetic Meridian
(iv) The process of limiting the flow of magnetic fields between two locations by separating them with a barrier made of conductive material.	D. Dip needle
(v) The process of disturbing the domain of an atom in magnetized material.	E. Temporary magnets
	F. Magnetic shielding
	G. Electromagnet
	H. Demagnetization
	I. Geographic Meridian

SECTION B (60 Marks)
(Answer all questions in this section)

3. a). i) Define the term power and state its SI unit
ii) Convert the following into watts
a) 1Kw b) 1Mw c) 1hp
b) How much power is required to accelerate a 1000kg car from rest to 26.7M/S in 8 second?
4. a) Why it is advised to fasten a seat belt when travelling in a car. Explain by using Newton's First law of motion.
b) A 4kg ball travelling at 6M/S to the right collides head on with a 6kg ball travelling at 4m/s the left. If the collision is perfectly elastic What are the velocities of two balls after collision?
5. a) State the laws of refraction.
b) When a ray of light is travel from air to glass, the angle of is 30° . If the refractive index of the glass is 1.5 Determine the angle of incidence.
6. a) Define the term scalar and give three examples.
b) Find the resultant Force, F, when two forces 9N and 15N act on an object with an angle of 60° between them.
7. a) state Pascal's principle of pressure
b) What are the two factors that affect the liquid pressure?
c) Calculate the area of an object if the pressure exerted is 0.2N/M^2 and its force is 2N.
8. a). i) Outline two Equilibrium conditions.
ii) State any three (3) factors upon which these equilibrium conditions depend on.
b) The diagram below shows a rod balanced on a fulcrum. How far from the point must the 50g mass be placed for the system to be in equilibrium.



SECTION C (25 Marks)

(Answer all questions in this section)

9. (a) Explain the terms friction, normal reaction, and viscosity.
(b) Mention three laws of friction.
(c) A block of mass 270kg is pulled along the horizontal surface. If the coefficient of kinetic friction between the block and surface is 0.4. What is the force acting on the block as it slides?
(d) An aluminum block of mass 2.1kg rest on a steel plat form a horizontal force of 15N is applied to the block
(i) Given that the coefficient of limiting friction between the two surface is 0.6, block will move?
(ii) If it moves, what will be its acceleration? If coefficient of kinetic friction of kinetic friction of aluminum is 0.47.
10. (a) State the law of conservation of energy
(b) A pendulum bob of mass 50g is pulled aside to a vertical height of 20cm from the horizontal and then released. Find
(i) The maximum potential energy of the bob.
(ii) The maximum speed of the bob
(iii) The K.E of the bob when it is at a height of 8cm from the horizontal
(c) Name one device which change
(i) Sound energy into electric energy
(ii) Electrical energy into heat energy